

Exercise 1: Multiple Linear Regression

Using this dataset:

```
data = {  
    'area': [50, 65, 70, 85, 100, 120, 140, 160, 180, 200],  
    'rooms': [2, 2, 3, 3, 3, 4, 4, 5, 5, 6],  
    'distance_km': [12, 10, 9, 8, 7, 6, 5, 4, 3, 2], # distance from town center  
    'price': [14500, 19000, 21200, 25800, 30500, 36500, 42500, 48800, 55200, 61500]  
}
```

Tasks:

1. Train a LinearRegression model using **all three features** (area, rooms, distance_km) to predict price
2. Print the intercept and all three coefficients
3. Predict the price of a house with: 115m², 4 rooms, 6km from town center
4. Calculate MSE, MAE, and RMSE on the test set
5. Which feature has the biggest coefficient? What does that tell you about its influence on price?